

# Staff Payroll Demonstration

**Title:** Staff Payroll System using Inheritance and Runtime Polymorphism

**Objective/Aim:**

To design and implement a Java program to calculate and display the salary structure of permanent and temporary staff using inheritance, constructors, method overriding, and dynamic method dispatch.

**System Requirements :**

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**Hardware**

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1]
- Minimum 4 GB / 8 GB RAM[cite: 1]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

**Software**

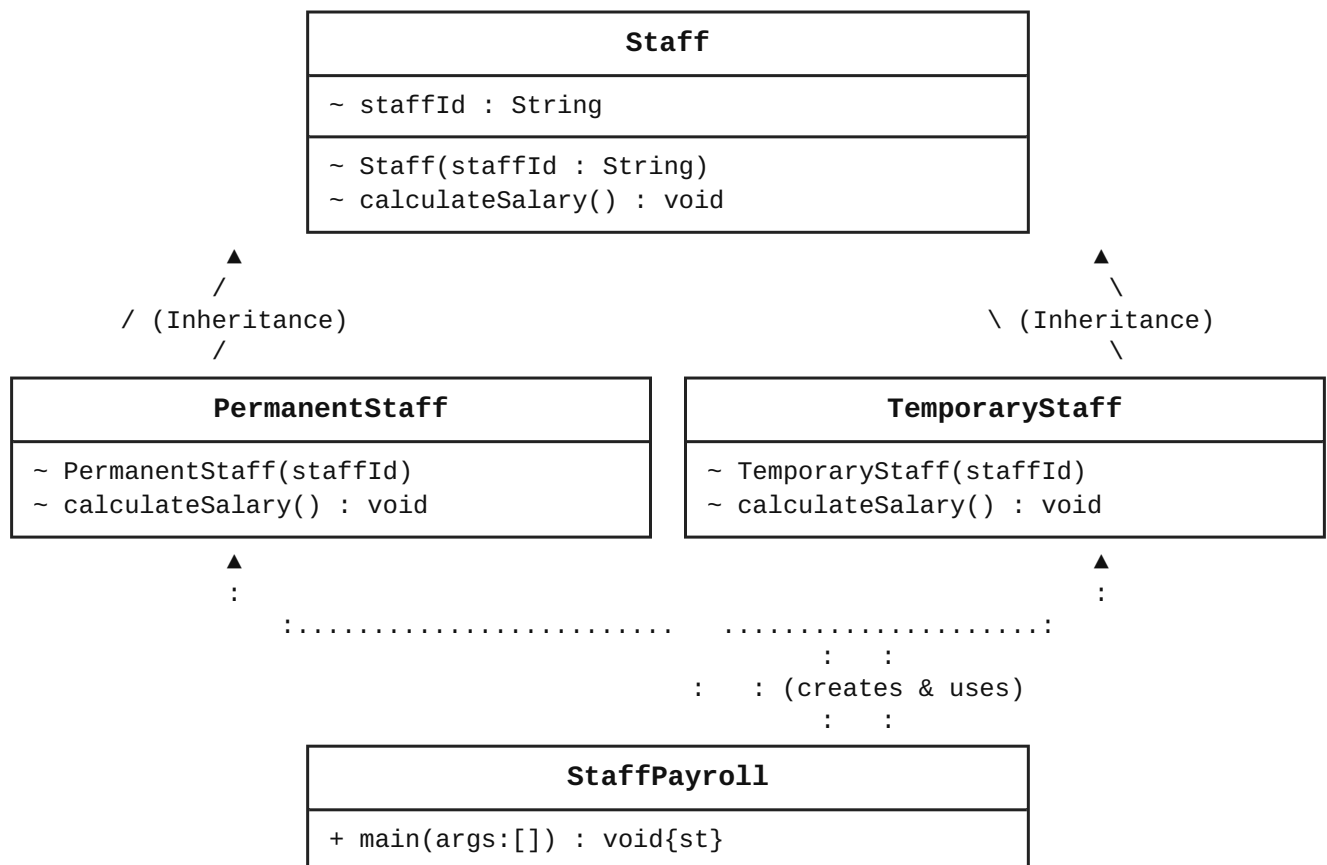
- Operating System: Windows/Linux/macOS[cite: 1]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1]
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code[cite: 1]
- Command Prompt/Terminal[cite: 1]

**Algorithm / Logic Description:**

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1. Create a superclass named `Staff` with attribute `staffId`.
2. Define a constructor to initialize `staffId` and a base method `calculateSalary()`.
3. Create a derived class `PermanentStaff` extending `Staff`.
4. Override `calculateSalary()` in `PermanentStaff` to calculate gross salary with basic (28000), allowance (12000), and travel allowance (4000).
5. Create a derived class `TemporaryStaff` extending `Staff`.
6. Override `calculateSalary()` in `TemporaryStaff` to calculate gross salary with basic (15000), allowance (2000), and travel allowance (2500).
7. Create the `StaffPayroll` class containing the `main()` method.
8. Initialize a staff identifier (e.g., "P101").
9. Use conditional statements (`startsWith("P")` or `startsWith("T")`) to instantiate the appropriate subclass through a superclass reference variable (`Staff worker`).
10. Invoke `calculateSalary()` through the reference to demonstrate runtime polymorphism.
11. Stop the program[cite: 2].

## UML class Diagram



**Relationship:** PermanentStaff and TemporaryStaff extend the base class Staff. The driver class StaffPayroll instantiates and invokes them dynamically via superclass references.

## Source Code:

```
// Parent class
class Staff {

    String staffId;

    // Constructor
    Staff(String staffId) {
        this.staffId = staffId;
    }

    // Salary method
    void calculateSalary() {
        System.out.println("Employee Salary Information:");
    }
}

// Permanent Employee class
class PermanentStaff extends Staff {

    // Constructor
    PermanentStaff(String staffId) {
```

```

        super(staffId);
    }

    // Overriding calculateSalary()
    @Override
    void calculateSalary() {

        int basic = 28000;
        int allowance = 12000;
        int travel = 4000;

        int grossSalary = basic + allowance + travel;

        System.out.println("Salary Information:");
        System.out.println("Basic Salary : " + basic);
        System.out.println("Allowance      : " + allowance);
        System.out.println("Travel Allow.: " + travel);
        System.out.println("Gross Salary : " + grossSalary);
    }
}

```

```

// Temporary Employee class
class TemporaryStaff extends Staff {

    // Constructor
    TemporaryStaff(String staffId) {
        super(staffId);
    }

    // Overriding calculateSalary()
    @Override
    void calculateSalary() {

        int basic = 15000;
        int allowance = 2000;
        int travel = 2500;

        int grossSalary = basic + allowance + travel;

        System.out.println("Salary Information:");
        System.out.println("Basic Salary : " + basic);
        System.out.println("Allowance      : " + allowance);
        System.out.println("Travel Allow.: " + travel);
        System.out.println("Gross Salary : " + grossSalary);
    }
}

```

```

// Main class
public class StaffPayroll {

    public static void main(String args[]) {

        String staffId = "P101";

        // Identify employee type using ID
        if (staffId.startsWith("P")) {

```

```
        Staff worker = new PermanentStaff(staffId);

        worker.calculateSalary();

    }
    else if (staffId.startsWith("T")) {

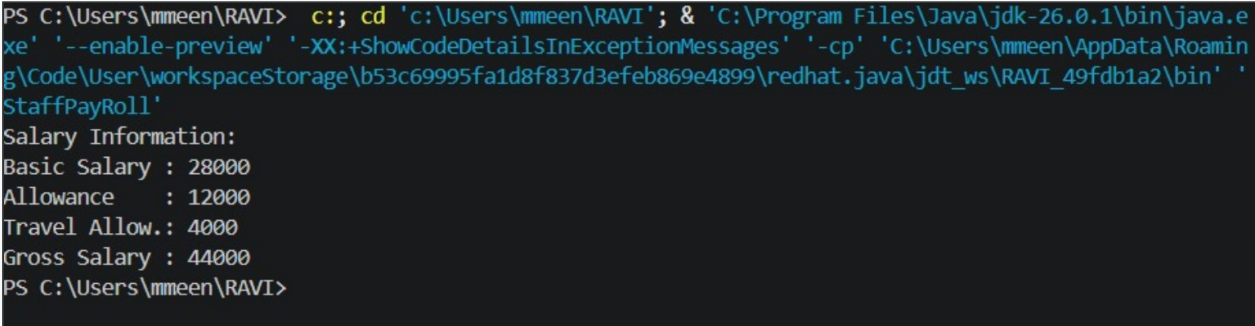
        Staff worker = new TemporaryStaff(staffId);

        worker.calculateSalary();

    }
}
```

## Output Screenshot:

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```
PS C:\Users\mmmeen\RAVI> c::; cd 'c:\Users\mmmeen\RAVI'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\mmmeen\AppData\Roaming\Code\User\workspaceStorage\b53c69995fa1d8f837d3efeb869e4899\redhat.java\jdt_ws\RAVI_49fdb1a2\bin' 'StaffPayRoll'
Salary Information:
Basic Salary : 28000
Allowance    : 12000
Travel Allow.: 4000
Gross Salary : 44000
PS C:\Users\mmmeen\RAVI>
```